22-inch LCDs
The New Entry Level?

Dell SP 2208 WFP
Add a little gloss to your life

The worst thing about Dell's new 22-inch Ultrasharp is the fact that it's based around a TN LCD panel. We have just one last grouse and then we're done, and then some of us are going out to see if we can buy one. The silver bezel that some love, and others hate, and the entire fit and finish, exudes quality. The stand is finished in piano black, and adds a striking contrast to all the silver. A 2 megapixel Web camera has been cleverly integrated into what has to be one of the slimmest bezels ever to grace an LCD to date. No ugly bulges, no protrusions, and the SP 2208 WFP looks like it's seen the inside of a wind tunnel during its design. We do feel Dell goofed up with a glossy LCD panel rather than the matte ones we're used to seeing for one huge reason—reflection. It's like looking into a mirror, especially with any sort of light source in the background, and we found this quite annoying during our tests.

Other than this one niggle and fact that a silver bezel isn't the best idea for any LCD (read glare), the SP 2208 WFP boasted of an intuitive menu system and good, easy to use buttons—very ergonomic. The integrated camera has excellent quality. This one features an HDMI port as well—superb connectivity options.

Considering the fact that a similar discrete webcam could cost as much as Rs 2,500, this monitor is a superb deal at Rs 15,386, and the price alone makes a very strong point to purchase it. And that's before you even consider that it's a good performer. We had issues with the Intensity Range Check in DisplayMate where we missed out on a lot of the darker grey squares which were indistinguishable from the black background. In the reverse text tests we had issues with bright green text on a black and grey background—the text just wasn't visible. Colour intensity was very good for a TN panel, especially the colour green, which is very easy to discern. This monitor also did well in the 256-shade ramp test, where even slight gradations were noticeable—of course it cannot produce the entire intensity gamut like an S-IPS panel, but then we didn't expect it to.

It does well at games, and the very contrast-sensitive *F.E.A.R.* looked quite good—deep shadows with good variations in relative lightness and darkness of a screen. In HD movies you will notice its slightly deficient contrast, but it's nothing out of the ordinary for a TN panel—and we've seen much worse.

Aside from a couple of design flaws and the mirror-like panel, there's nothing wrong with this monitor, and you'll hardly find a better conglomerate of features and performance at the price. If you're looking for a monitor within the range of Rs 16,000, you'd have to be daft not to consider the SP2208 WFP as a serious option.

How We Tested

We divided our test candidates into five categories on the basis of screen size.
Features: Besides the theoretical parameters which we mention we awarded points for presence of add-ons like Web cameras, speakers, USB ports, and memory card readers. We awarded points according to the connectivity options available. Aside from this we also looked at certain features like the ability to pivot and swivel, or the presence of portrait mode and height adjustment. We graded the looks of the LCD and the intuitiveness of the menu system and the button layout. We rated the quality of moving parts as well.

Operating System	Windows XP Professional
CPU	Core 2 Duo E6700 (2.66 GHz)
Motherboard	Gigabyte DQ6 (P965)
RAM	Kingston HyperX 512x2 DDR2 800 MHz
Graphics Card	NVIDIA GeForce 8800GTX
Hard Drive	Seagate 7200.9 250GB SATA

Benchmark	Tests	What we looked for
DisplayMate Video Edition	Geometry Tests	Point Shape and visibility, Pixel sharpness, Line resolution, shape integrity. This basically tests the ability of a monitor to work with shapes like cones, spheres, cubes, and even 2D shapes. Useful for professional users. Professionals will want to look here.
	Contrast and greyscale Tests	Checks the ability to produce subtle variations in intensity of a single shade, and the ability to display purest blacks, and whitest whites.
	Colour Tests	Tests the colour rendition ability of an LCD. Uses a number of tests like colour purity tests, reverse text tests, shade intensity and colour ramp tests for the same. Animation, image editing and rendering professionals look at these values.
Movie Tests	Pixel persistence, colour and contrast	A mean value is taken for all these parameters, pixel persistence is a measure of response time, while colour and contrast are necessary parameters for a proper movie experience
	Overall experience	Here we look at the type of panel, whether it reflects a lot of light, the bezel colour and its interference with the image onscreen, and the aspect ratio—which also affects the overall experience.
Game Tests	F.E.A.R	Very dark and atmospheric game, with a lot of weapon, environment and special effects including bullet time and particle effects that are still the best in the business, very good to test contrast ratios of monitors as it trips most of them up with its ability to produce lifelike shadows, and their interaction with light sources which is very realistic. A good benchmark for gamers to base decisions on.
	Oblivion	Unbelievably detailed environments, with some great foilage and overall texturing. Also showcases transparency effects very well, and has some of the best colours seen in any game. Usually trips up monitors with texture detailing on floors, sand and other environment shaders, and even bloom and transparency effects. Gamers look here and decide.
Viewable Angle	Text and Movie	Here we look at the ability to view the monitor from angles other than straight on, this reveals how much distortion of the image and and its colours will occur if you happen to be sitting at an angle. Important for those people looking at an HTPC monitor, simply because with a number of people watching, say, a movie, it becomes impossible for each person to be directly in front of the monitor.

HP L2208w
When matte met flat

If you're not one for loud statements then you'll appreciate the laid back, even bland design on this one. A matte finished body extends to the bezel—black is the best colour for an LCD bezel, and if it's dull enough to refrain from reflecting ambient light so much the better. The stand looks cool with a silver and black colour tone. However the swivel action on the stand is quite hard, and a couple of days of, err, swivelling didn't loosen things much. The control buttons aren't backlit or colour indicated, and fiddling with them in dim lighting is a hit and miss affair. The slight indentation in the centre of the stand is quite useful for keeping a few utilitarian knickknacks like pencils, erasers or screwdrivers.

DisplayMate's Intensity Range Check showed a lot of indiscernible grey squares—this bespeaks a poor contrast ratio. We faced no issues with the 16-shade ramp test, but in the 256-colour ramp test we had issues with the variation in intensity of colours green and grey. The movie tests revealed slightly less colour rendition and contrast in most HD clips (compared to the Dell and Samsung monitors of the same size). In *F.E.A.R* we had issues with contrast where the enemies in dark khaki uniforms were virtually invisible when in shadowy areas. This is a serious put off for any gamer especially when playing dark, atmospheric games.

All in all at Rs 18,500 the L2208w is a costly proposition, both on your wallet and your visual senses. It's a great looking monitor that corrects some of the flaws of Dell's SP 2208 WFP design, but also detracts a lot on the performance front. With the exception of the extreme rarity of you being a diehard HP fan, give this one a miss.

AOC 2217 Pwc
Strictly mediocre stuff, this...

It's a good looking monitor but the fact that it integrates speakers doesn't excuse the rather broad bezel. The piano finish is quality. The stand is very big for a monitor that doesn't have any kind of height adjust, and it managed to fool us that it did have this feature till we used it. Three USB ports add some connectivity options. The contrast ratio isn't too good as a couple of quick DisplayMate tests revealed. This dismal performance continued

in the 16 shade intensity test where we couldn't discern any difference between two or three of the most intense shades for each colour. Due to its lack of a proper colour rendition, contrast calibrating was a chore. While it performed decently at games like *F.E.A.R* (no worse than the HP L2208w) it didn't do as well at movies and we recommend you avoid this panel for any kind of serious multimedia PC.

At Rs 16,000 the AOC 2217Pwc is slightly costly, but it does have a serious competitor in Dell, who manages to outdo it on all fronts, including price. One position is its decent performance in the geometry tests along with its competitive price (compared to the HP) which make it a decent buy for an office monitor.

ViewSonic VX2255WM
White n Bright...

A piano white finish is somewhat a rarity in this colour prejudiced LCD world where black

is a defacto standard. The three little birds (ViewSonic's trademark) stand out in visible relief and detail from all the white, another perk for deviating from black. Speaking about the bezel, it isn't as narrow as we'd have liked—especially considering the space hogging buttons have been moved to the side. The trim on the lower bezel looks a little out of place and flimsy.

In DisplayMate the contrast was immediately better than the HP and AOC 22-inch offerings, although not as good as the Samsung and the Dell monitors. No serious issues in the colour tests but the rendition of green isn't good enough. In *F.E.A.R* we noticed a decent contrast ratio and good particle effects but both Dell and Samsung are better. *Oblivion* is much more forgiving and the VX2255WM makes a very good showing in games with dimly lit scenes, something where its TN panel bearing compatriots also performed reasonably well on. Movies show good colour but mediocre contrast, especially noticeable in darker scenes and with dim indoor lighting. At Rs 15,500, the VX2255WM is a decent offering—good enough for multimedia, games, and office use. However it doesn't really excel at any of these applications, and there are better options available.